

**Introduction to Research Methodology****Introduction:****I. WHAT IS RESEARCH?**

Research is a scientific and systematic search for pertinent information on a specific topic.

According to Clifford Woody research comprises of defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organizing and evaluating data; and reaching conclusions; and finally, carefully testing the conclusions to determine whether they fit the formulating hypothesis.

In summary, research is the search for knowledge, using objective and systematic methods to find solution to a problem.

**II. OBJECTIVES OF RESEARCH:**

1. To gain familiarity with the phenomenon
2. To describe the characteristics of a particular individual, situation or a group
3. To determine the frequency with which something occurs or with which it is associated with something else
4. To test hypothesis of a casual relationship between variables

**III. TYPES OF RESEARCH:**

1. Descriptive vs. Analytical
2. Applied vs. Fundamental
3. Quantitative vs. Qualitative
4. Conceptual vs. Empirical
5. Some other types of research

**1. Descriptive vs. Analytical:**

a. Descriptive research includes surveys and fact-finding enquiries of different kinds.

Researcher has no control over the variables, he can report only what has happened or what is happening.

*Example:* Frequency of shopping, preference of people. etc.,

b. In analytical research, researcher has to use the facts or information already available and analyze these to make critical evaluation.

Examples:

1. How to reduce number of complaints made by customers
2. How absentee rate among employees be reduced

**2. Applied vs. Fundamental:**

a. Applied research aims at finding a solution for an immediate problem (social or economical or political problems) facing a society or an industrial organization.

Example: **Vaccine for corona virus.**

b. Fundamental research is mainly concerned with generalizations and with the formulation of the theory.

Example: Natural concepts, Pure mathematics, etc.,

### 3. Quantitative vs. Qualitative

a. Quantitative research is based on the quantitative measurements of some characteristics.

Example: How many people prefer design 'A'

b. Qualitative research is concerned with qualitative phenomenon relating to quality or kind.

Example: Reasons for human behavior. (how people think about subject or institution), 'why people prefer design A'.

### 4. Conceptual vs. Empirical

a. Conceptual research is related to some abstract ideas or theory. It is generally used by philosophers and thinkers to develop new concepts or reinterpret existing ones.

b. Empirical research relies on experience or observation alone, without due regard for system and theory. It is data-base research, coming up with conclusions which are capable of being verified by observation or experiment. It is also called as experimental type of research.

### 5. Some other types of Research:

1. One time research (**the research is confined to a single time-period**)

or longitudinal research (**the research is carried on over several time periods**).

2. Field setting research or laboratory research or simulation research (Ex: **creating prototype and simulating**)

3. Clinical or Diagnostic research (Such research follows *case-study* methods or in-depth approaches to reach the basic causal relations, medicines for new viruses)

4. Exploratory research (*without experiment only theory*)

5. **Historical research** - utilizes historical sources like documents, remains, etc. to study events or ideas of the past, including the philosophy of persons

6. **Conclusion oriented or Decision oriented research** - A researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes.

## IV. RESEARCH APPROCHES:

### a. Quantitative approach:

1. Inferential approach: *It uses surveys to study characteristics or relationships of population.*

Determine the features of north Indians and south Indians and inference is drawn to know that the population has same features.

2. Experimental approach: *Experimental approach involves in manipulating variables to observe their effects (tensile test: Load vs deformation)*

3. Simulation Approach: *Simulation approach involves the construction of a artificial environment within which the relevant information or data can be generated. (Creating prototype model)*

- b. **Qualitative approach** to research is concerned with subjective assessment of attitudes, opinions and behavior. Generally, these techniques focus on group interviews, projective techniques, etc.,

## V. RESEARCH METHODS vs. METHODOLOGY

1. Research methods or techniques refer to the methods researchers use in performing research operations.
2. Research methods refer to the methods used by researchers to study the problem. These methods are considered analytical tools in research, as they help researchers arrive at a solution for a problem.

*They are divided into 3 groups.*

- ✓ *Data collection methods*
- ✓ *Statistical techniques for establishing relationships between data and unknowns.*
- ✓ *Methods for evaluating the accuracy of results.*

Research methodology is a systematic approach for solving research problems, focusing on the steps taken by researchers to study their problem and the logic behind them.

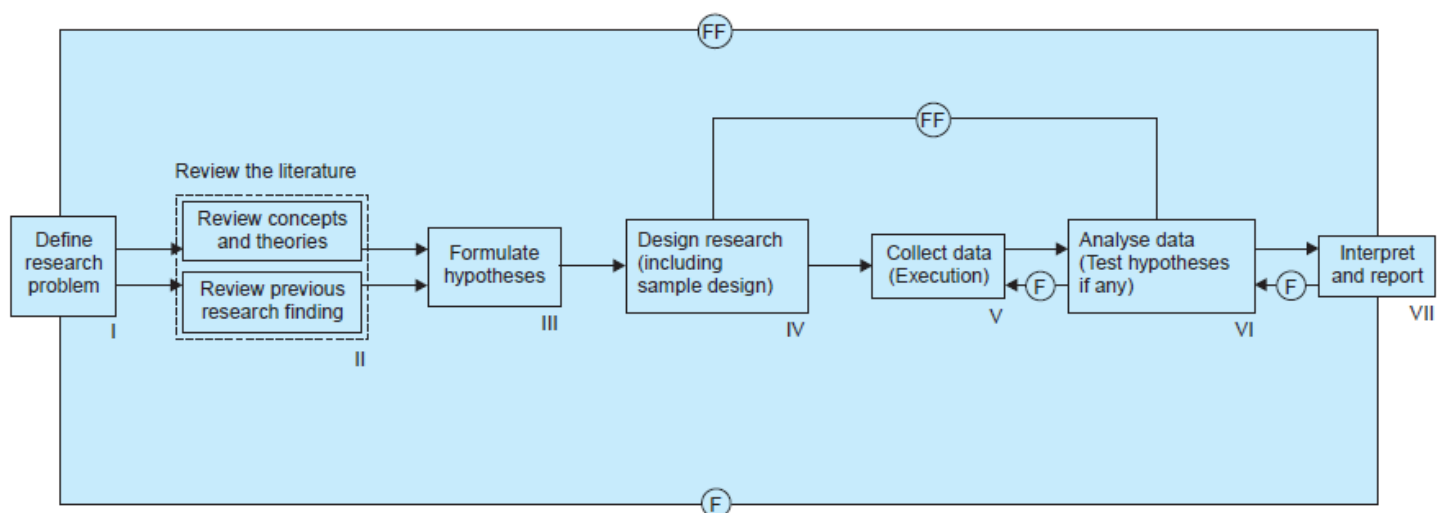
Researchers must understand not only the research methods and techniques but also the assumptions underlying them.

They must also determine which techniques are relevant and why, and how to determine which ones are applicable to specific problems.

This approach is crucial for researchers, as it may differ from problem to problem.

## VI. RESEARCH PROCESS

RESEARCH PROCESS IN FLOW CHART



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)

(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

- Formulating the research problem
- Extensive literature survey
- Developing Hypothesis
- Preparing the research design
- Determining sample design
- Collecting the data
- Execution of the project
- Analysis of data
- Hypothesis testing
- Generalizations and interpretation
- Preparation of the report or presentation of the results.

**1. Formulating the research problem:** Researchers first identify their research problem, determining the general area of interest. Essentially two steps are involved in formulating the research problem.

a. Understanding the problem thoroughly (Discussion)

b. Rephrasing the same into meaningful terms from an analytical point of view. (Resources)

Understanding a problem requires discussing it with colleagues or experts. In academic institutions, researchers can seek guidance from experienced guides, who can help narrow down the problem in operational terms.

## **2. Extensive literature survey**

The researcher should conduct a comprehensive literature survey, utilizing various sources such as abstracting and indexing journals, academic journals, conference proceedings, government reports, and books, with careful consideration of previous similar studies and the aid of a good library.

## **3. Developing Hypothesis**

- ✓ The working hypothesis is a tentative assumption made after extensive literature survey, providing the focal point for research and affecting data analysis.
- ✓ The process involves discussing the problem with colleagues and experts, examining available data for trends, reviewing similar studies, and conducting exploratory personal investigations, including limited-scale field interviews, to gain insight into the practical aspects of the problem.

## **4. Preparing the research design:**

A research design is a conceptual structure that facilitates efficient research by collecting relevant evidence with minimal effort, time, and money. It is crucial for research purposes, which can be categorized into exploration, description, diagnosis, and experimentation. A flexible design is appropriate for exploration purposes, while a suitable design minimizes bias and maximizes data reliability for accurate descriptions or associations between variables.

Research designs include experimental and non-experimental hypothesis testing, with options including informal or formal designs.

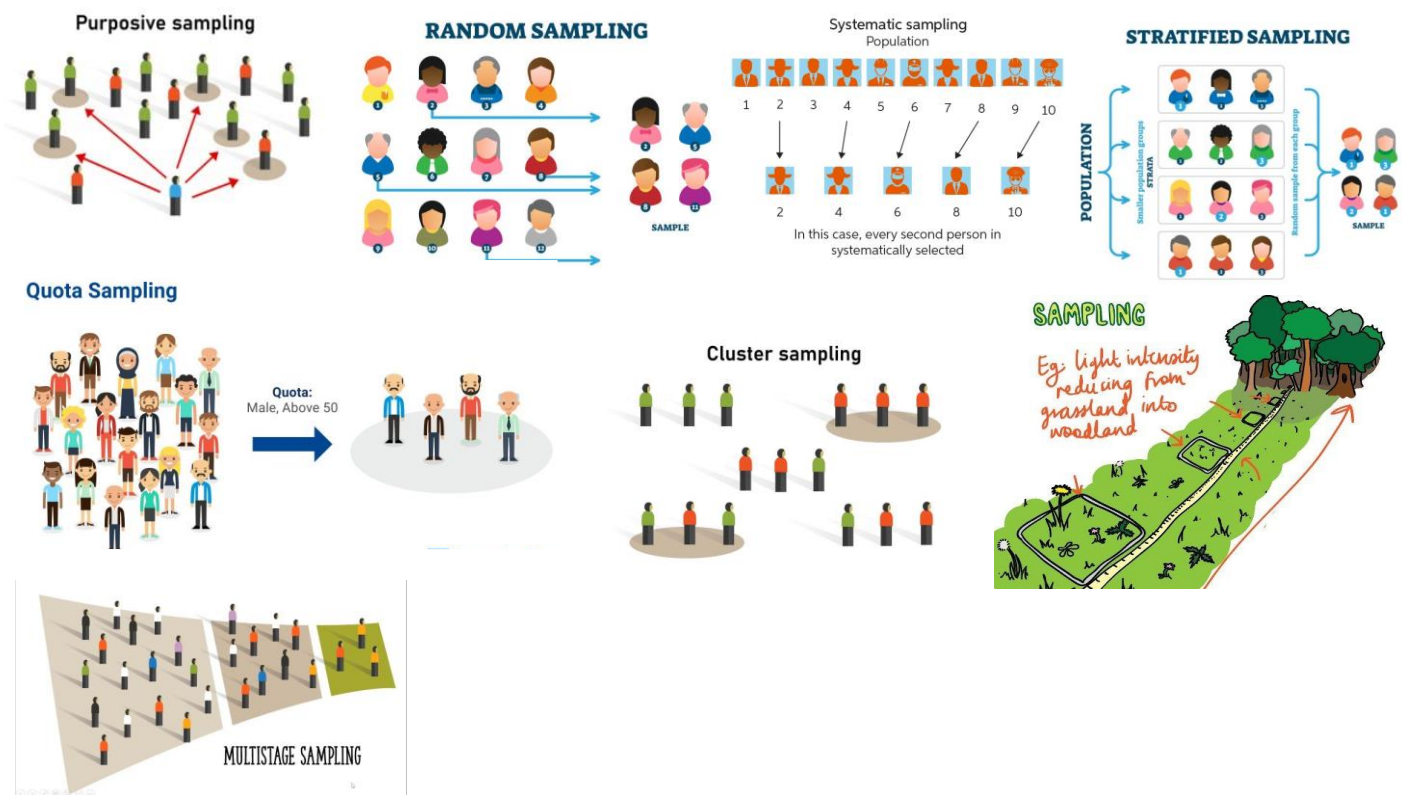
Researchers must choose the appropriate design based on factors like information sources, researcher and staff skills, organization, time, and research cost.

## 5. Determining sample design:

- ✓ All the items under consideration in any field of inquiry constitute a 'universe' or 'population'.
- ✓ A census inquiry is a comprehensive enumeration of all items in a population, aiming to eliminate chance and achieve highest accuracy.

Different sample designs are:

- a. Deliberate sampling or purposive or non-probability sampling
- b. Simple random sampling
- c. Systematic sampling
- d. Stratified sampling
- e. Quota sampling
- f. Cluster sampling
- g. Area sampling
- h. Multi-stage sampling
- i. Sequential sampling



- a. Deliberate sampling, also known as purposive or non-probability sampling, involves selecting specific units of the universe for a sample. Convenience sampling selects population elements based on accessibility, while judgement sampling uses the researcher's judgement to select items

representative of the population. Both methods can lead to biased results, especially when the population is not homogeneous. Judgement sampling is commonly used in qualitative research.

b. **Simple random sampling**

*This type of sampling is also known as chance sampling or probability sampling where each and every item in the population has an equal chance of inclusion in the sample and each one of the possible samples, in case of finite universe, has the same probability of being selected.*

- c. **Systematic sampling** is a method of selecting every element in a list, often using random numbers to select the unit. This method is useful when a sampling frame is available in the form of a list, starting with a random point and selecting every element until the desired number is reached.
- d. **Stratified sampling** is a technique used to draw a representative sample from a non-homogeneous population by stratifying it into nonoverlapping subpopulations or strata and selecting items from each stratum using simple random sampling.
- e. **Quota sampling** is a non-probability sampling method where interviewers are given quotas to fill from different strata, with the selection of items left to the interviewer's judgement, as random samples are often more expensive.
- f. **Cluster sampling** is a method of grouping a population into clusters, rather than individual elements, for inclusion in a sample. For example, a departmental store could group its 15,000 credit card holders into 100 clusters of 150 card holders each, randomly selecting three clusters for the sample. This approach can increase efficiency in field work, particularly in personal interviews.
- g. **Area sampling** divides a large geographical area into smaller clusters, randomly selecting units from these smaller areas. This method is useful when population lists are unavailable and enhances efficiency in field interviews by allowing multiple interviews at each location.
- h. **Multi-stage sampling** is a technique for large-scale inquiries, starting with selecting primary units like states, districts, towns, and families within towns, and then applying random-sampling at all stages for multi-stage random sampling.
- i. **Sequential sampling** is a complex sample design where the final sample size is determined based on information gathered during the survey, typically used for statistical quality control.

**6. Collecting the data:** Observation, personal interview (structured procedure), Telephone interview, Questionnaires.

**7. Execution of the project:** should ensure a systematic and timely execution of the project

**8. Analysis of data:** through categorization, coding, tabulation, and drawing statistical inferences

**9. Hypothesis testing:** Chi square, t-test, and F-test, are used to test hypotheses

**10. Generalizations and interpretation:** process of explaining findings based on a theory, can trigger new questions and lead to further research.

**11. Preparation of the report or presentation of the results:** prepare a report with a careful layout, including preliminary pages, main text, and end matter.

## **VII. CRITERIA FOR GOOD RESEARCH**

- ✓ Scientific research should be clearly defined, detailed, objective, and reportable.
- ✓ It should be well-planned, with flaws in design being reported and their effects on findings.
- ✓ Data analysis should be adequate, valid, and reliable. Conclusions should be based on the data, and experienced researchers with a good reputation and integrity can provide greater confidence in the research. This ensures continuity and continuity in the research process.

## **VIII. QUALITIES OF GOOD RESEARCH**

- Good research is *systematic* means research is structured with specified steps to be taken in a specified sequence in accordance with the well-defined set of rules.
- It is *logical*, guided by induction and deduction, making research more meaningful in decision-making contexts.
- It is *empirical*, dealing with concrete data from real situations, providing external validity to results.
- It is *replicable*, allowing results to be verified through replication, providing a sound basis for decisions.

## **IX. Defining the Research Problem**

A researcher must find the problem and formulate it so that it becomes susceptible to research.

Like a medical doctor, a researcher must *examine all the symptoms* concerning a problem before he can *diagnose correctly*.

## **X. Components of Research Problem**

- There must be an individual or a group which has some difficulty or the problem.
- There must be some objective(s) to be attained at. If one wants nothing, one cannot have a problem.
- There must be alternative means (or the courses of action) for obtaining the objectives one wishes to attain. This means that there must be *at least two means* available to a researcher for if he has no choice of means, he cannot have a problem.
- There must remain some doubt in the mind of a researcher with regard to the selection of alternatives. This means that research must answer the question concerning the relative efficiency of the possible alternatives.
- There must be some environment(s) to which the difficulty pertains.

## **XI. SELECTING THE PROBLEM**

- Subject which is overdone should not be normally chosen, for it will be a difficult task to throw any new light in such a case.

- Controversial subject should not become the choice of an average researcher.
- Too narrow or too vague (unclear) problems should be avoided.
- The subject selected for research should be familiar and feasible so that the related research material or sources of research are within one's reach.
- The importance of the subject, the qualifications and the training of a researcher, the costs involved, the time factor are few other criteria that must also be considered in selecting a problem.
- The selection of a problem must be preceded by a preliminary study. (if new)

## **XII. TECHNIQUE INVOLVED IN DEFINING A PROBLEM**

1. **Statement of the problem in a general way:** The researcher can himself state the problem or he can seek the guidance of the guide or the subject expert in accomplishing this task. Often, the guide puts forth the problem in general terms, and it is then up to the researcher to narrow it down and phrase the problem in operational terms.
2. **Understanding the nature of the problem:** The best way of understanding the problem is to discuss it with those who first raised it in order to find out how the problem originally came about and with what objectives in view. If the researcher has stated the problem himself, he should consider once again all those points that induced him to make a general statement concerning the problem.
3. **Surveying the available literature:** the researcher must be well-conversant with relevant theories in the field, reports and records as also all other relevant literature. He must devote sufficient time in reviewing of research already undertaken on related problems.
4. **Developing the ideas through discussions:** researcher must discuss his problem with his colleagues and others who have enough experience in the same area or in working on similar problems.
5. **Rephrasing the research problem:** Once the nature of the problem has been clearly understood, the environment (within which the problem has got to be studied) has been defined, discussions over the problem have taken place and the available literature has been surveyed and examined, rephrasing the problem into analytical or operational terms is not a difficult task. Through rephrasing, the researcher puts the research problem in as specific terms as possible so that it may become operationally viable.

In addition to what has been stated above, the following points must also be observed while defining a research problem:

- Technical terms and words or phrases, with special meanings used in the statement of the problem, should be clearly defined.
- Basic assumptions or postulates (if any) relating to the research problem should be clearly stated.
- A straight-forward statement of the value of the investigation (i.e., the criteria for the selection of the problem) should be provided.
- The suitability of the time-period and the sources of data available must also be considered by the researcher in defining the problem.



- The scope of the investigation or the limits within which the problem is to be studied must be mentioned explicitly in defining a research problem.

### **XIII. NECESSITY OF DEFINING THE PROBLEM:**

Quite often we all hear that a problem clearly stated is a problem half solved. This statement signifies the need for defining a research problem. The problem to be investigated must be defined unambiguously for that will help to discriminate relevant data from the irrelevant ones. A proper definition of research problem will enable the researcher to be on the track whereas an ill-defined problem may create hurdles. It is only on careful detailing the research problem that we can work out the research design and can smoothly carry on all the consequential steps involved while doing research.